UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

PHYSICS – IMPORTANT FORMULAE & CONSTANTS

GRADE 9 ADVANCED | TERM 1 | 2026–2027

MODULE 1 | PHYSICS TOOLKIT — MEASUREMENT & MATHEMATICS

MEAN OF REPEATED READINGS

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

x̄ = mean; x = reading (same unit); n = number of readings (unitless).

PERCENTAGE ERROR

$$\frac{|x_{\text{measured}} - x_{\text{accepted}}|}{|x_{\text{accepted}}|} \times 100\%$$

Compare a measured value with a non-zero accepted value. Both values use the same unit.

PERCENTAGE UNCERTAINTY

$$\frac{\text{absolute uncertainty}}{|\text{measured value}|} \times 100\%$$

Absolute uncertainty has the same unit as the measurement; result is a percentage.

DENSITY

$$\rho = \frac{m}{V}$$

ρ = density (kg/m³); m = mass (kg); V = volume (m³).

RECTANGULAR AREA

$$A = l \times w$$

A = area (m²); l = length (m); w = width (m).

RECTANGULAR VOLUME

$$V = l \times w \times h$$

V = volume (m³); l, w, h = length, width and height (m).

MODULE 1 | SEVEN SI BASE QUANTITIES

BASE QUANTITY	SI BASE UNIT	SYMBOL
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

MODULE 1 | SI PREFIXES — MULTIPLIERS

PREFIX	SYMBOL	MULTIPLIER
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}

MODULE 1 | SIGNIFICANT FIGURES — RULES & OPERATIONS

RULE	EXAMPLE / APPLICATION
Non-zero digits	347 has 3 significant figures.
Zeros between non-zero digits	1007 has 4 significant figures.
Leading zeros	0.0045 has 2 significant figures.
Trailing decimal zeros	2.300 has 4 significant figures.
Addition and subtraction	Use the fewest decimal places: 12.11 + 0.3 = 12.4.
Multiplication and division	Use the fewest significant figures: 2.5 × 3.42 = 8.6.
Exact counts / defined conversions	Do not limit significant figures. Round only the final answer.

MODULE 1 | GRAPHING DATA & STRAIGHT LINES

GRADIENT / SLOPE

$$\text{slope} = \frac{\Delta y}{\Delta x}$$

Change in vertical quantity ÷ change in horizontal quantity. Unit = y-unit per x-unit.

STRAIGHT-LINE EQUATION

$$y = mx + b$$

m = constant gradient; b = y-intercept. Here m is gradient, not mass.

Recognize the shape: linear: $y \propto x$; quadratic: $y \propto x^2$; inverse: $y \propto 1/x$. Plot the independent variable horizontally and the dependent variable vertically.

MODULE 2 | REPRESENTING MOTION — POSITION & DISPLACEMENT

DISPLACEMENT

$$\Delta x = x_f - x_i$$

Δx = displacement (m); x_f = final position (m); x_i = initial position (m).

ELAPSED TIME

$$\Delta t = t_f - t_i$$

Δt = elapsed time (s); t_f = final time (s); t_i = initial time (s).

TOTAL DISTANCE

$$d = |\Delta x_1| + |\Delta x_2| + \cdots$$

d = total path length (m). Add each segment without a direction sign.

AVERAGE SPEED

$$v_{\text{speed}} = \frac{d}{\Delta t}$$

Average speed (m/s) = total distance (m) ÷ total elapsed time (s).

AVERAGE VELOCITY

$$\bar{v} = \frac{\Delta x}{\Delta t}$$

$\bar{v}$ = average velocity (m/s). Use signed displacement, not total distance.

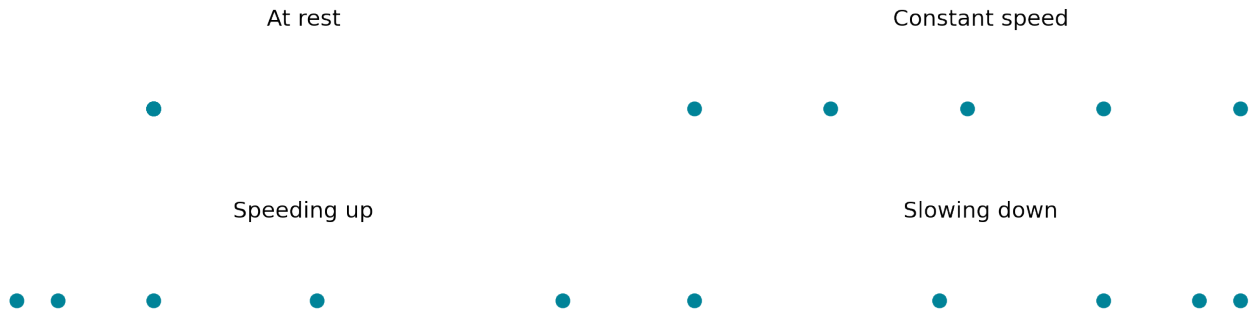
CONSTANT VELOCITY

$$x_f = x_i + v\Delta t$$

x_f, x_i = position (m); v = constant velocity (m/s); Δt = elapsed time (s).

MODULE 2 | MOTION DIAGRAMS

Dots mark positions at **equal time intervals**. The arrow gives direction. Larger gaps mean greater speed.



Vectors: displacement, velocity, acceleration and force have direction. **Scalars:** distance, speed, mass, time, work and energy.

MODULE 3 | ACCELERATED MOTION — EQUATIONS

Use the four motion equations below only when acceleration is **constant**. Keep one positive direction throughout.

AVERAGE ACCELERATION

$$\bar{a} = \frac{v_f - v_i}{\Delta t}$$

$\bar{a}$ = average acceleration (m/s²); v_i, v_f = initial, final velocity (m/s).

VELOCITY–TIME EQUATION

$$v_f = v_i + a\Delta t$$

Use when initial velocity, acceleration and elapsed time are known.

DISPLACEMENT–TIME EQUATION

$$\Delta x = v_i\Delta t + \frac{1}{2}a(\Delta t)^2$$

Δx = displacement (m); a = constant acceleration (m/s²); Δt = time interval (s).

EQUATION WITHOUT TIME

$$v_f^2 = v_i^2 + 2a\Delta x$$

Use when time is not given. $\Delta x = x_f - x_i$; all velocities must include signs.

AVERAGE-VELOCITY METHOD

$$\Delta x = \frac{v_i + v_f}{2}\Delta t$$

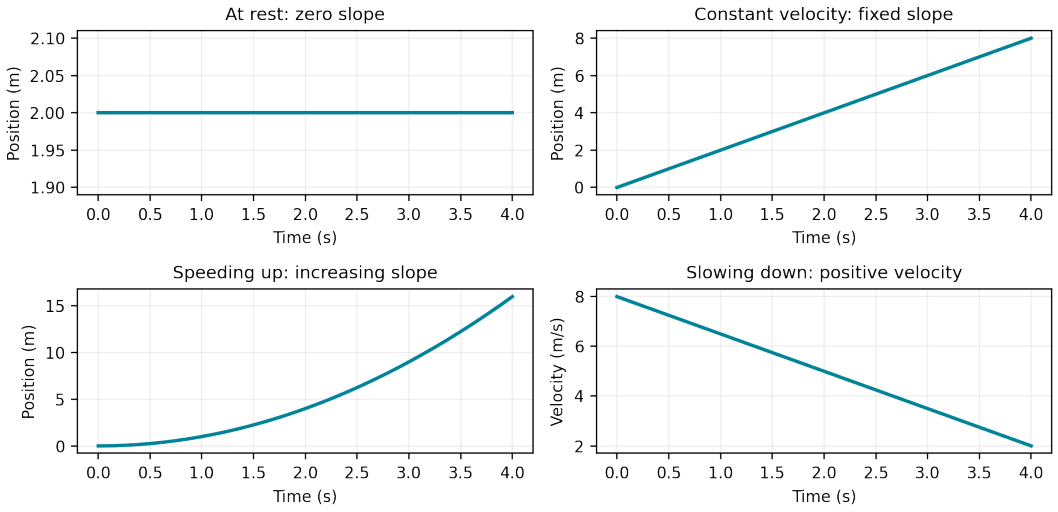
For constant acceleration only: average velocity is the mean of initial and final velocities.

FINAL POSITION

$$x_f = x_i + v_i\Delta t + \frac{1}{2}a(\Delta t)^2$$

Position = initial position + displacement. All position terms use metres.

MODULES 2–3 | MOTION GRAPHS — VISUAL REFERENCE



Speeding up: velocity and acceleration have the same sign. **Slowing down:** they have opposite signs. Negative acceleration alone does not mean slowing down.

MODULES 2–3 | SLOPES & AREAS

POSITION–TIME SLOPE

$$\bar{v} = \frac{\Delta x}{\Delta t}$$

Secant-line slope = average velocity. Tangent-line slope = instantaneous velocity (m/s).

VELOCITY–TIME SLOPE

$$\bar{a} = \frac{\Delta v}{\Delta t}$$

Secant-line slope = average acceleration. Tangent-line slope = instantaneous acceleration (m/s²).

VELOCITY–TIME AREA

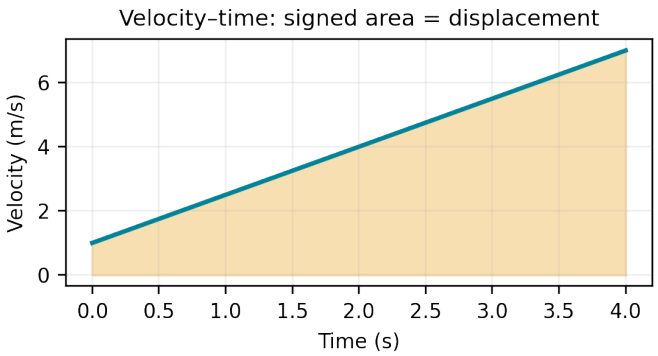
$$\Delta x = \text{signed area}$$

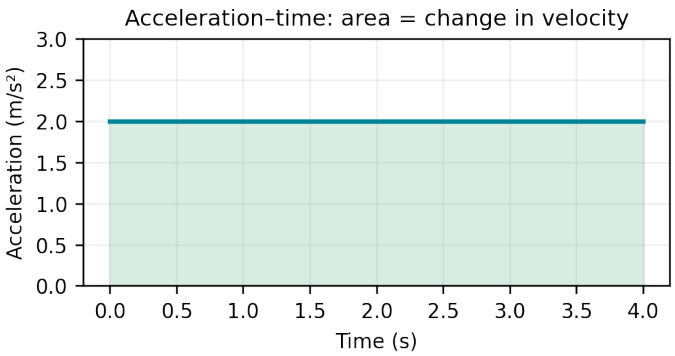
Area above the time axis is positive; area below it is negative. Unit: m.

ACCELERATION–TIME AREA

$$\Delta v = \text{signed area}$$

Δv = change in velocity (m/s). For a speed–time graph, area = total distance (m).





MODULE 3 | FREE FALL, CONSTANTS & CONVERSIONS

FREE-FALL ACCELERATION

$$g = 9.8\ \text{m/s}^2$$

g is the magnitude. If upward is positive, a = −g; if downward is positive, a = +g.

VERTICAL MOTION

$$v_f^2 = v_i^2 + 2a\Delta y$$

Replace Δx with Δy in the constant-acceleration equations. At the top, v = 0 but a ≠ 0.

CONVERSION / CHECK	USE
Length / time	1 km = 10^3 m; 1 cm = 10^{-2} m; 1 mm = 10^{-3} m; 1 min = 60 s.
Speed	km/h → m/s: divide by 3.6; m/s → km/h: multiply by 3.6.
Graph area	Rectangle = base × height; triangle = ½ base × height; trapezium = ½(a + b) × height.
Final check	Δ means final minus initial. State units and direction; round only at the end.